Dhwanil Ranpura

San José, CA | dranpura1907@gmail.com | linkedin.com/in/dhwanil-ranpura | github.com/dhwanil1907 | dhwanil-portfolio.vercel.app

Skills

Languages: Python, SQL, R, TypeScript, JavaScript
ML & Statistics: XGBoost, scikit-learn, Logistic Regression, SHAP, ARIMA, Anomaly Detection, Feature Engineering
Data: pandas, NumPy, PostgreSQL, DuckDB, PySpark, ETL, star schema / dimensional modeling, data-quality checks
Engineering & BI: FastAPI, testing (pytest, Vitest), Git, Linux, Streamlit, Plotly, Power BI, Excel

Experience

Data Automation Intern Sep 2024 – Feb 2025
Language & Culture Worldwide (LCW) *Remote*

- Built a **Power BI** dashboard on team-level staffing, covering headcount and resource allocation
- Helped build a **Power Automate** flow that routes personnel assignments and sends automated emails

Projects

Home Credit Risk Scorer | *Source Code* | *Python, XGBoost, FastAPI, pytest* Apr 2026

- Built an end-to-end credit default scoring system on **307,511** applications across up to 5 joined tables; **XGBoost** reached **0.779 ROC-AUC** with **41.2% recall** and **27.7% precision** at an F1-tuned threshold on a held-out 20% test set
- Wrote **30 pytest tests**, including **6** FastAPI `TestClient` API tests for valid, missing and out-of-range inputs; reused the same feature-engineering functions in training and serving
- Served the model through a **FastAPI** scoring API with input validation; added **4** business policy rules after a **267%** debt-to-income applicant scored Low Risk, with **SHAP** explanations

Financial Data Warehouse Pipeline | *Source Code* | *Python, SQL, DuckDB, PySpark* May 2026

- Built an **ETL** pipeline loading **2,260,668** Lending Club loans from raw CSV into a **star-schema DuckDB** warehouse (1 fact, 4 dimensions) with idempotent loads, so reruns never duplicate records
- Added schema validation and data-quality checks at extract to protect **data integrity**, with **28 passing pytest tests**, including an integration test that runs the full pipeline into a real DuckDB file, and an optional **PySpark** path
- Wrote **13 SQL** queries with CTEs and window functions; default risk rises from **3.28%** (grade A) to **37.48%** (grade G), about 11x, against **11.88%** overall; includes a 3-month rolling default rate and cohort analysis

Product Analytics Pipeline | *Website* | *Python, SQL, PostgreSQL, Streamlit* Sep 2026

- Loaded **885,129** e-commerce events (407,283 users) into a 4-table **PostgreSQL** star schema with **7** documented cleaning decisions
- Wrote **SQL** funnel, cohort, retention and revenue-segment analyses: **5.24%** of viewers purchase, **90.9%** never add to cart, and the top 20% of buyers drive **67.77%** of revenue
- Recommended a 24-hour re-engagement nudge after product views over checkout optimization, shown in a live Streamlit dashboard

Proxy: Undo Layer for AI Agents | *TypeScript, MCP, Vitest* Oct 2026

- Wrote the core logic (compare, trace, rank, reversibility, undo planning) of an **MCP server** that catches mistaken agent bookings and plans their reversal; team of 3 at the **YC** Make Something Agents Want hackathon
- Owned the merchant modules and negotiate step that reverse bookings through self-serve cancel, merchant agents or email
- Built **34** eval scenarios; the system caught **39 of 44** mistakes, with the **5** misses documented

KPI Anomaly Detection System | *Source Code* | *Python, scikit-learn, SHAP, Streamlit* Oct 2026

- Built unsupervised **anomaly detection** on **1,950** trading days of AAPL data with **5** detectors and consensus voting
- Validated against **15** documented market events with no labels: **60%** Precision@5; top flags caught the Feb–Mar 2020 COVID crash

Leadership

President Dec 2025 – Present
Spartan Analytics

- Revived an inactive club, growing membership from **37 to 73**; lead **17** people across **6** cross-functional teams that collaborate on technical workshops, projects, corporate outreach, events, marketing and membership
- Designed and taught Python and SQL workshops (**30+** students); launched the first annual data competition (**30+** participants)

Treasurer Aug 2025 – Present
Responsible Computing Club

- Raised **$21,400** in club funding and built the budgeting system from scratch, allocating and tracking spend across **11** club teams
- Communicate the club's financial position to the exec board

Education

San José State University Expected Dec 2027
B.S. Data Science *San José, CA*

- **Best Overall Winner**, ASA DataFest 2025; 3rd of 91 teams, SJHacks 2025